

ABDULLAH BASIT

Embedded Hardware & PCB Design Engineer

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PROFILE

Embedded hardware engineer with 1.5+ years across space instrumentation, laser/electro-optics R&D and embedded systems. M1 Electrical Engineering & Control Systems at Université Grenoble Alpes (15.07/20, Mention Bien), entering M2 Wireless Integrated Circuits & Systems (2026–2027). Full hardware cycle: schematic capture and multilayer mixed-signal PCB layout in Altium, STM32 firmware, board bring-up and validation. Seeking a PFE / final-year internship (from early 2027) in embedded systems, PCB or RF hardware design.

EXPERIENCE

Embedded Hardware & PCB Design Intern — Centre Spatial Universitaire de Grenoble (CSUG)

Grenoble, France | 04/2026 – Present

- Designed the THC (thermal control) and EPS (power) boards of a PC/104 CubeSat payload stack: STM32, AD7124 24-bit PT100 acquisition, MOSFET heater drive, INA3221 power telemetry; CAN + LoRa (868 MHz) telemetry; SPI/I2C sensor interfaces — schematics, layout and hardware validation.
- Implemented MeshCore mesh-protocol packet encoding/decoding in C for RIOT OS and integrated it into the SX1303 LoRa gateway as a store-and-forward repeater — 58-check test suite, validated on Nucleo-L432KC + RAK5146; contribution submitted upstream as a pull request (thingsat/riot_modules).
- Built the balloon-campaign telemetry chain: sub-frame packet protocol and a Web Serial ground-station GUI with live plotting and command uplink.

Hardware Design Engineer — Laser & Electro Optical Solutions (LEOS)

Rawalpindi, Pakistan | 12/2024 – 08/2025

- Designed mixed-signal and high-frequency PCBs in Altium Designer with signal-integrity focus — differential-pair routing and layout optimisation using Saturn PCB Toolkit.
- Developed power-management and buck–boost converter circuits (LM5155/56/57/58, LM745) for detector and embedded systems.
- Integrated ADC/DAC and power-management ICs with SoM platforms and NVIDIA development kits; multilayer boards for laser/optical detector R&D (FAOD) and MPSoC sensor integration (AXU5EVB-P).

Embedded Systems Intern — Magnifo Engineering Solutions

Islamabad, Pakistan | 09/2024 – 10/2024

- Developed C/C++ firmware for microcontroller-based systems; peripheral interfacing over UART, SPI, I2C and GPIO.

Assistant Lab Engineer — Pakistan Institute of Engineering & Applied Sciences (PIEAS)

Islamabad, Pakistan | 09/2023 – 06/2024

- Programmed STM32 microcontrollers for embedded control; designed LabVIEW data-acquisition and measurement systems.

EDUCATION

M2 Wireless Integrated Circuits & Systems (WICS) — Université Grenoble Alpes

Grenoble, France | 2026 – 2027 (starting September 2026)

M1 Electrical Engineering & Control Systems (EEEA) — Université Grenoble Alpes

Grenoble, France | 2025 – 2026 — **15.07/20, Mention Bien** (transcript available on portfolio)

BS Electrical Engineering — Pakistan Institute of Engineering & Applied Sciences (PIEAS)

Islamabad, Pakistan | 2020 – 2024 — CGPA 3.51/4.00. Thesis: EEG/EMG biopotential acquisition PCB for brain–computer interfaces.

SELECTED PROJECTS

- MeshCore support for RIOT OS** — open-source mesh-protocol library, tests and SX1303 gateway integration; upstream PR: github.com/thingsat/riot_modules/pull/6.
- Predictive Maintenance Embedded AI** — TinyML motor-fault classifier on STM32L432KC (TensorFlow Lite / STM32Cube.AI), five fault classes, UART streaming and live dashboard.
- EEG/EMG BCI acquisition front-end** — INA128 instrumentation, driven-right-leg and active filtering; microvolt signal recovery driving an LSTM gesture classifier.

TECHNICAL SKILLS

PCB & CAD: Altium Designer, KiCad, multilayer mixed-signal layout, signal integrity, Saturn PCB Toolkit

Embedded: STM32 (HAL/LL), ESP32, C/C++, Python, FreeRTOS, RIOT OS, TinyML (STM32Cube.AI)

Protocols: CAN, SPI, I2C, UART/RS-485, LoRa/LoRaWAN, MQTT

Power & Analog: SMPS design (buck–boost), battery charge paths (BQ25798), instrumentation front-ends, PID control

RF & EDA: Cadence Virtuoso, Keysight ADS, LTspice, MATLAB/Simulink

Lab & Tools: Oscilloscope, logic analyzer, VNA, fine-pitch soldering/rework, Git, Linux

LANGUAGES

English (fluent) • Urdu (native) • French (intermediate)